Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

SCHOOL OF ELECTRONICS ENGINEERING

Fall Semester 2025-2026

BECE406E – FPGA Based System Design

TASK 1 – FSM (MEALY AND MOORE)

SANJAY ELAVARASAN KARTHIKEYAN

23BEC0503

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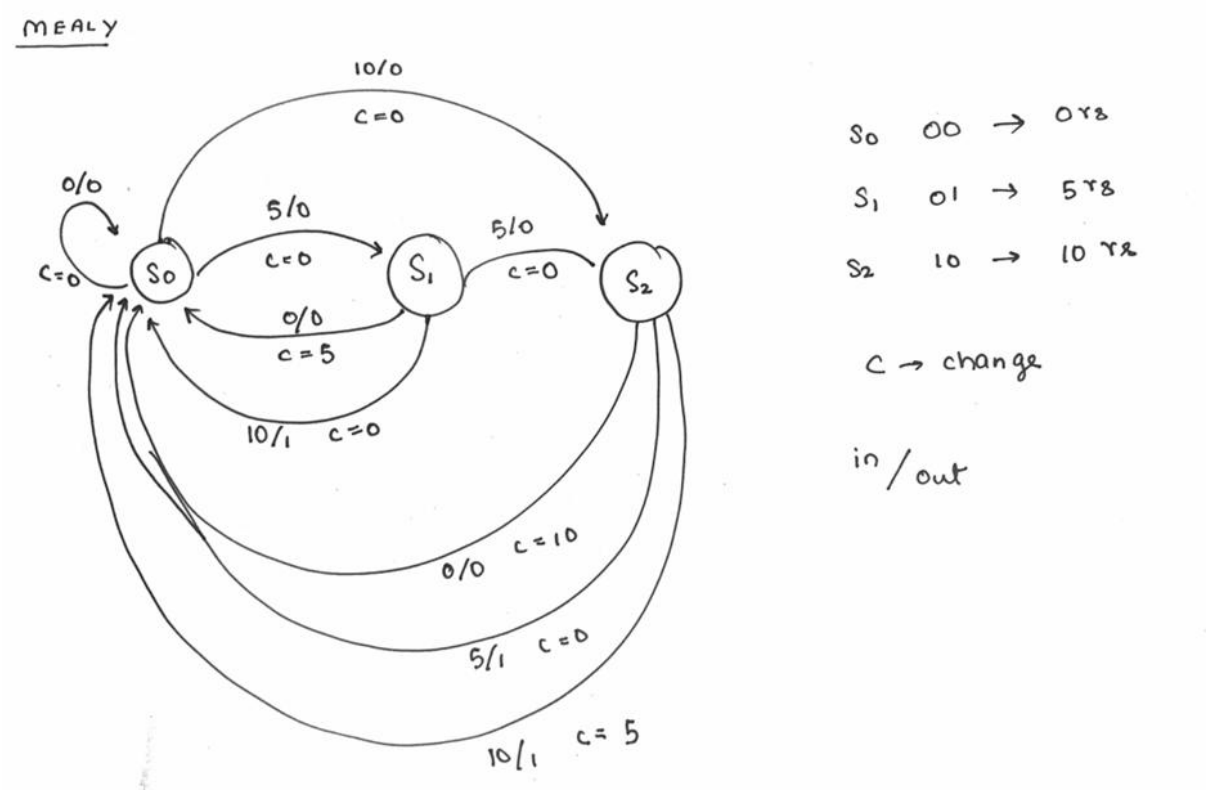
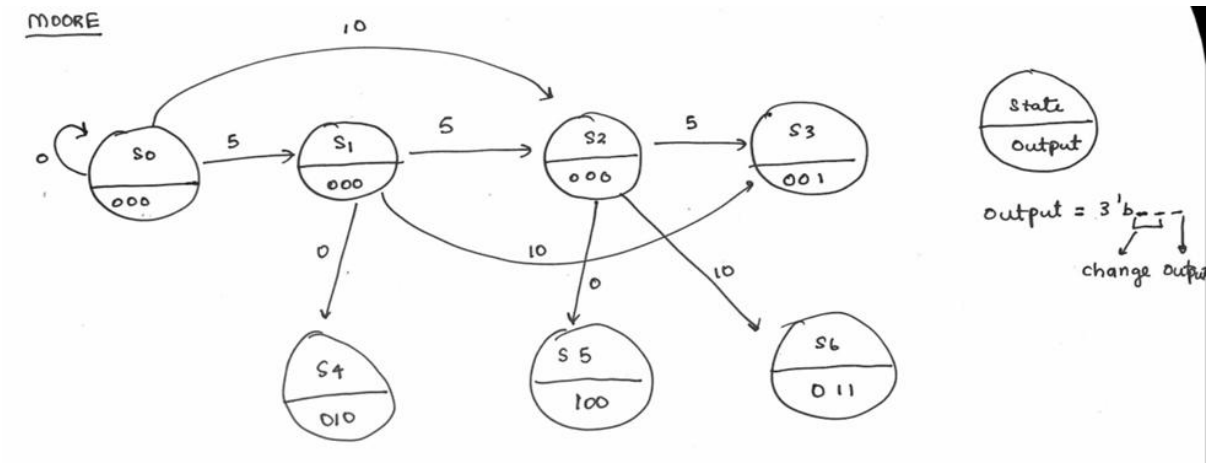
i) Design a Coke Vending machine using Mealy and Moore model which accepts Rs.5 and Rs.10 denominations only and deliver coke for Rs.15 and change if required.

MEALY:



MOORE:



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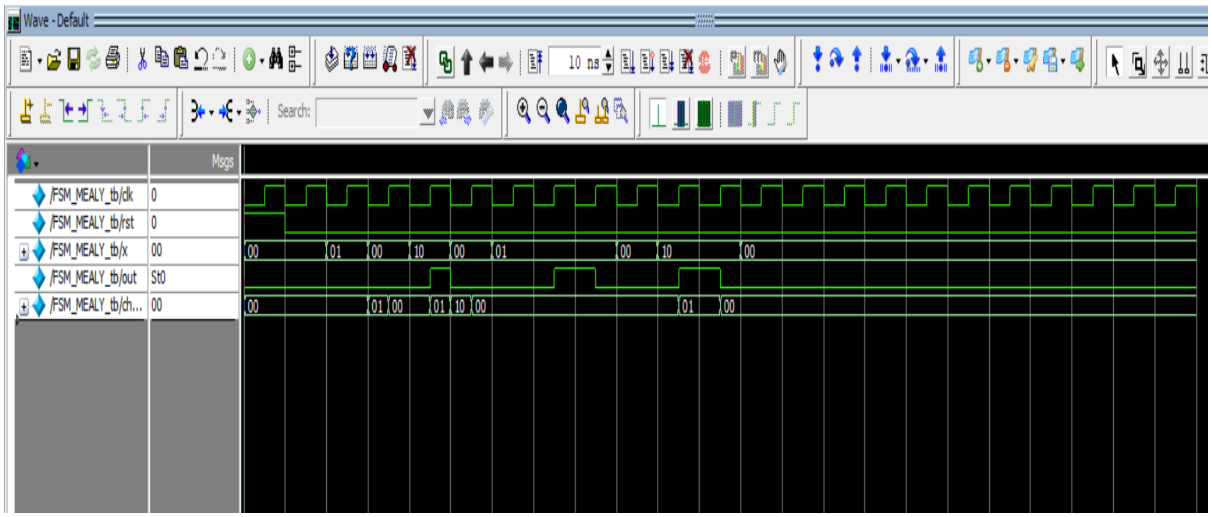
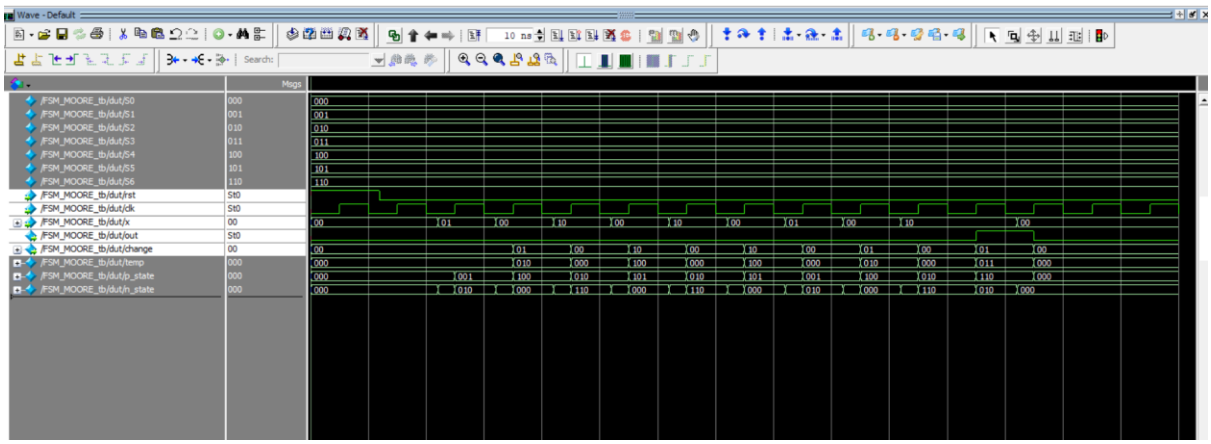
ii) Verify the functionality of coke vending machine using Mealy and Moore model using Modelsim.

MEALY :



MOORE:



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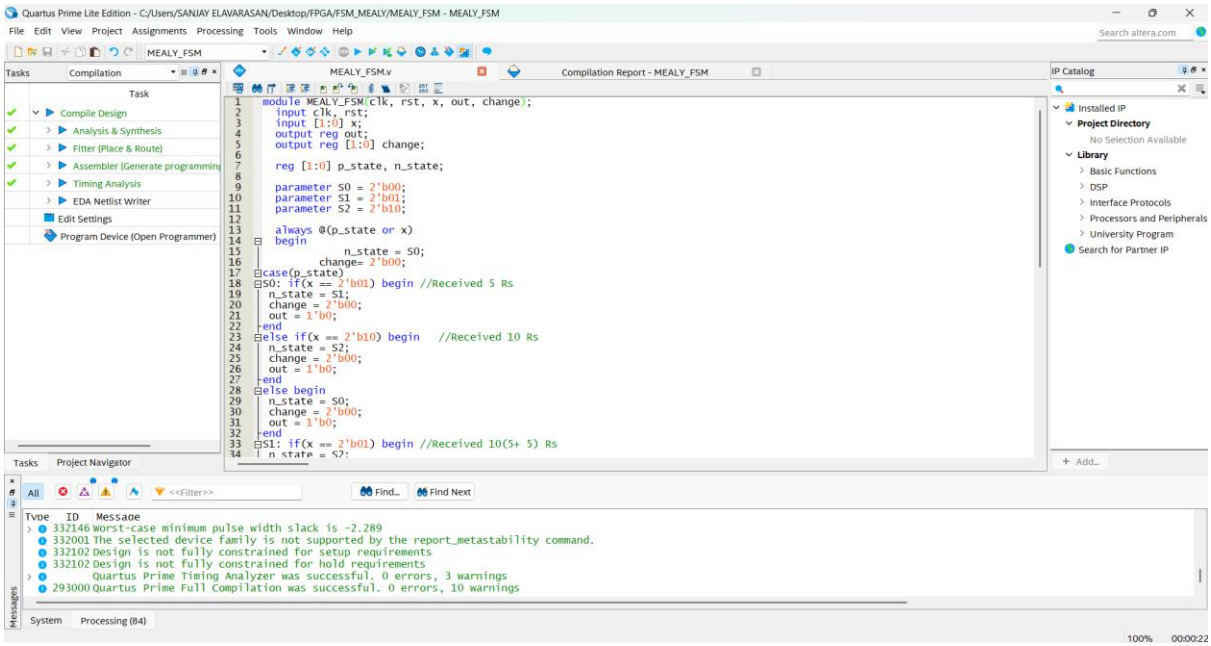
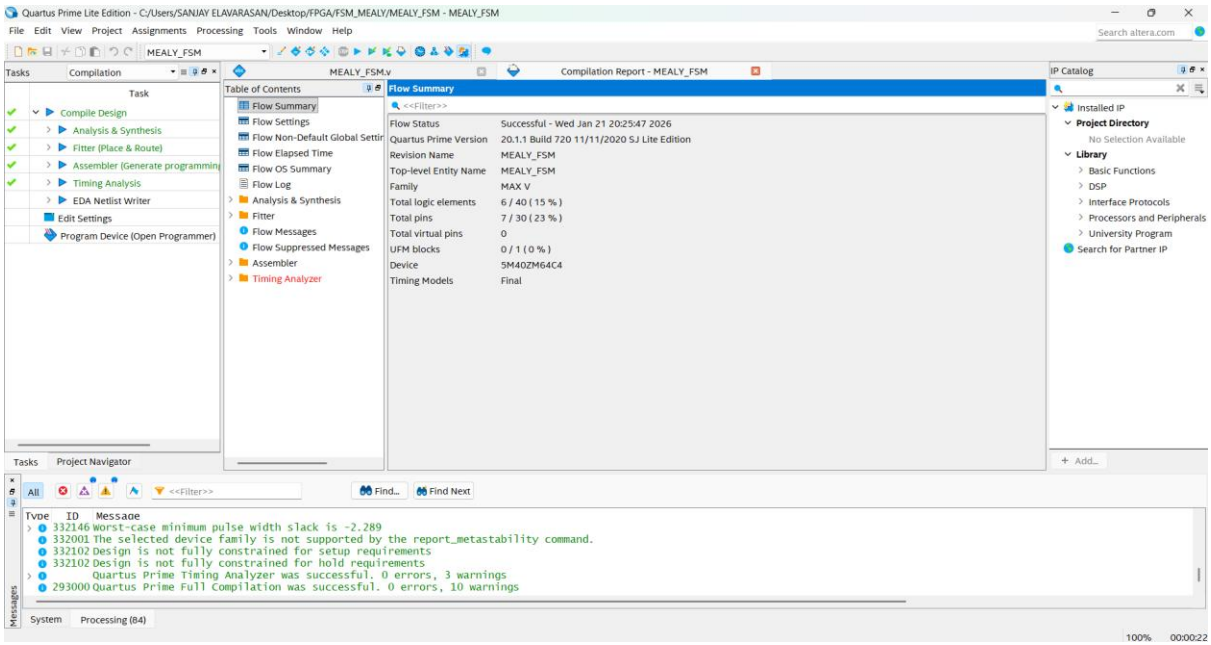
iii) Implement the Coke Vending machine using Mealy and Moore model in DE2-115 Board using Intel Quartus Prime.

MEALY:





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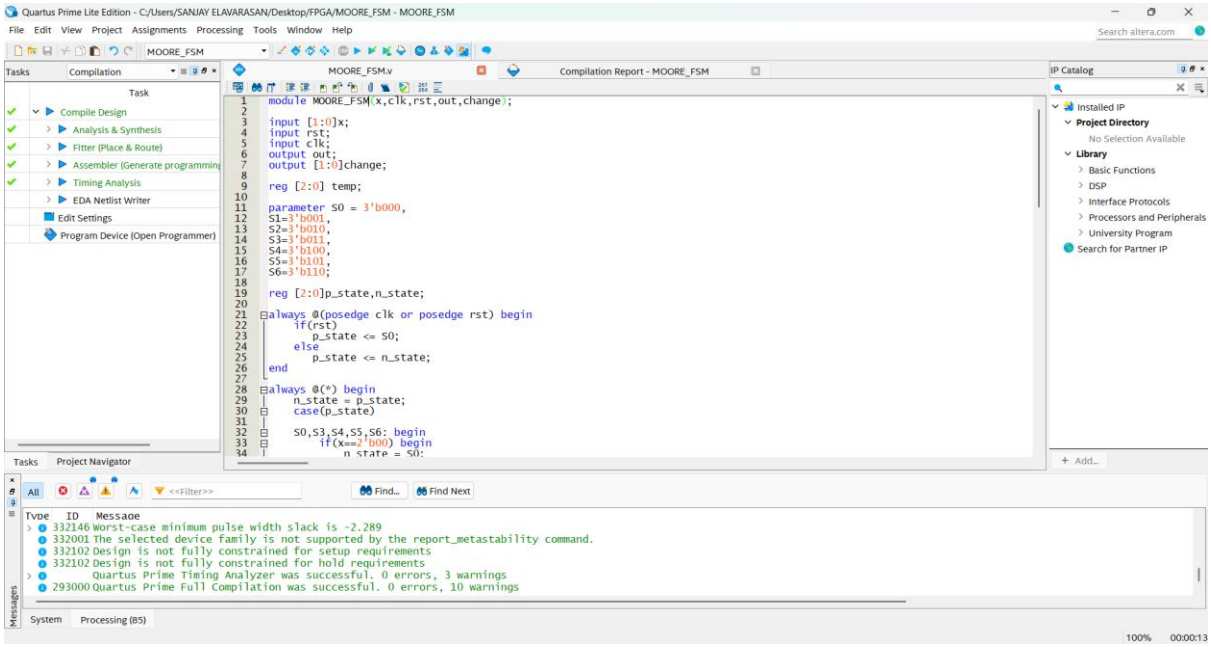
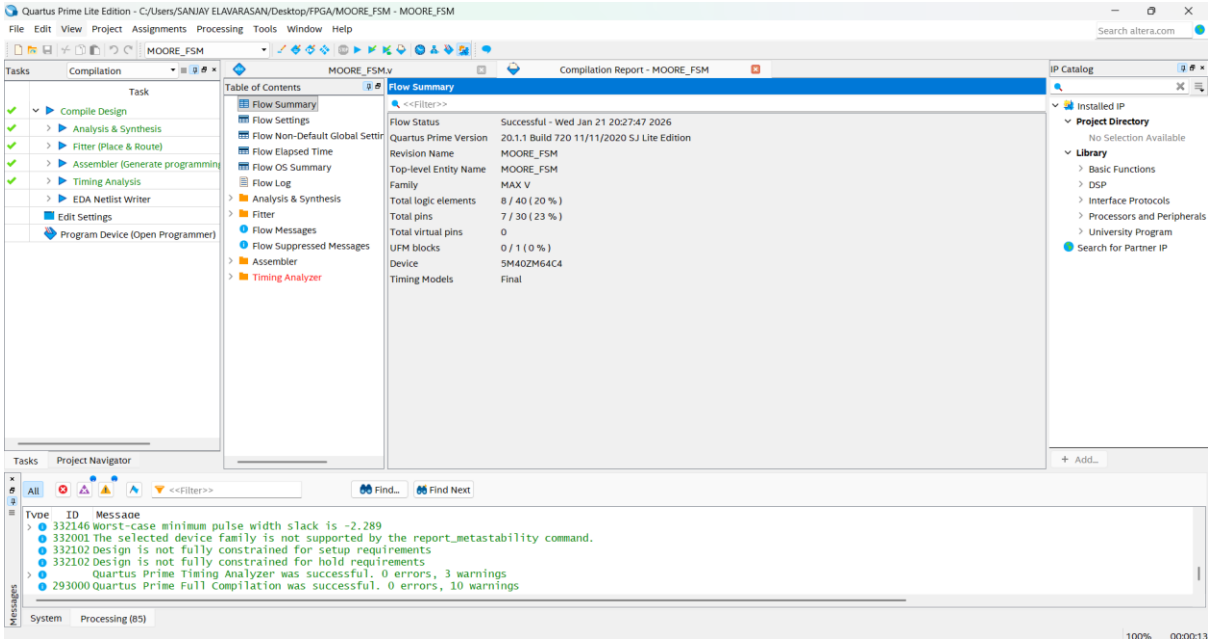
MOORE:





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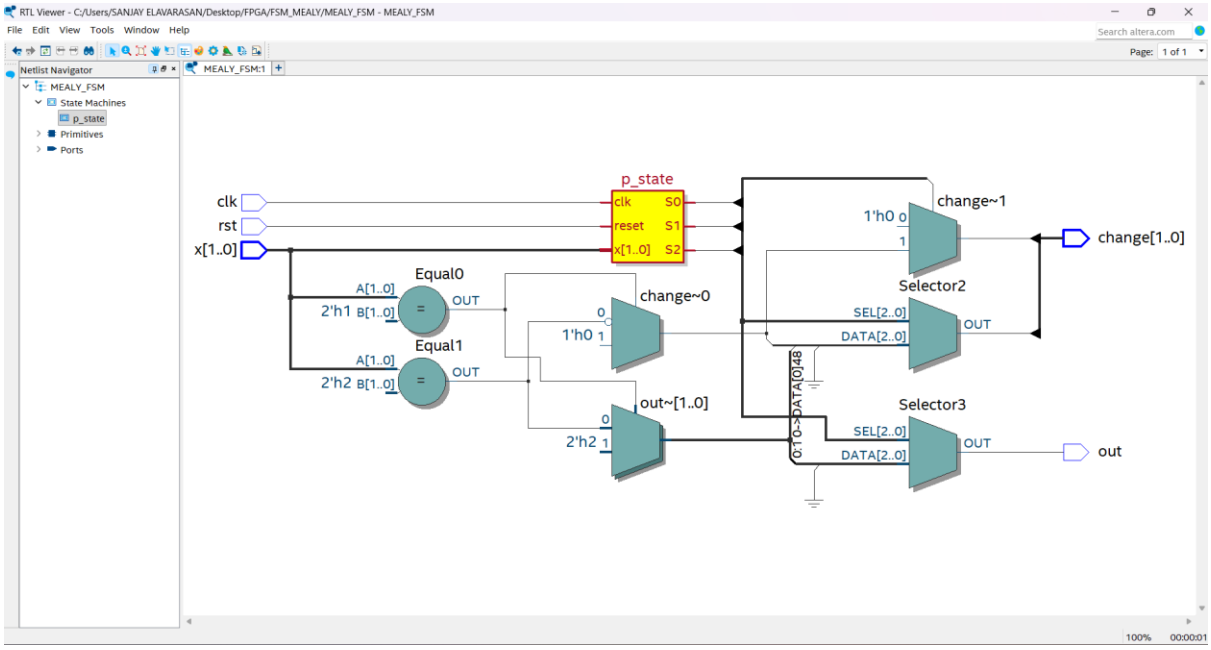
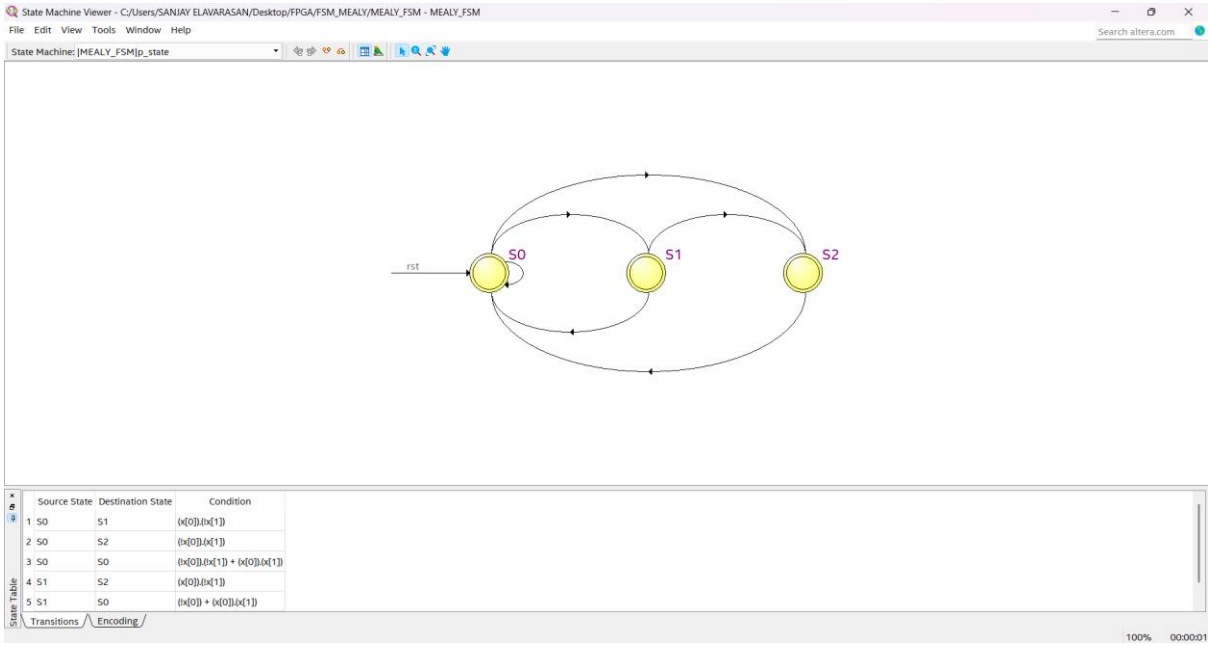
v) Attach the RTL view and Screenshots of code and outputs of Coke Vending machine.

RTL VIEW:

MEALY :





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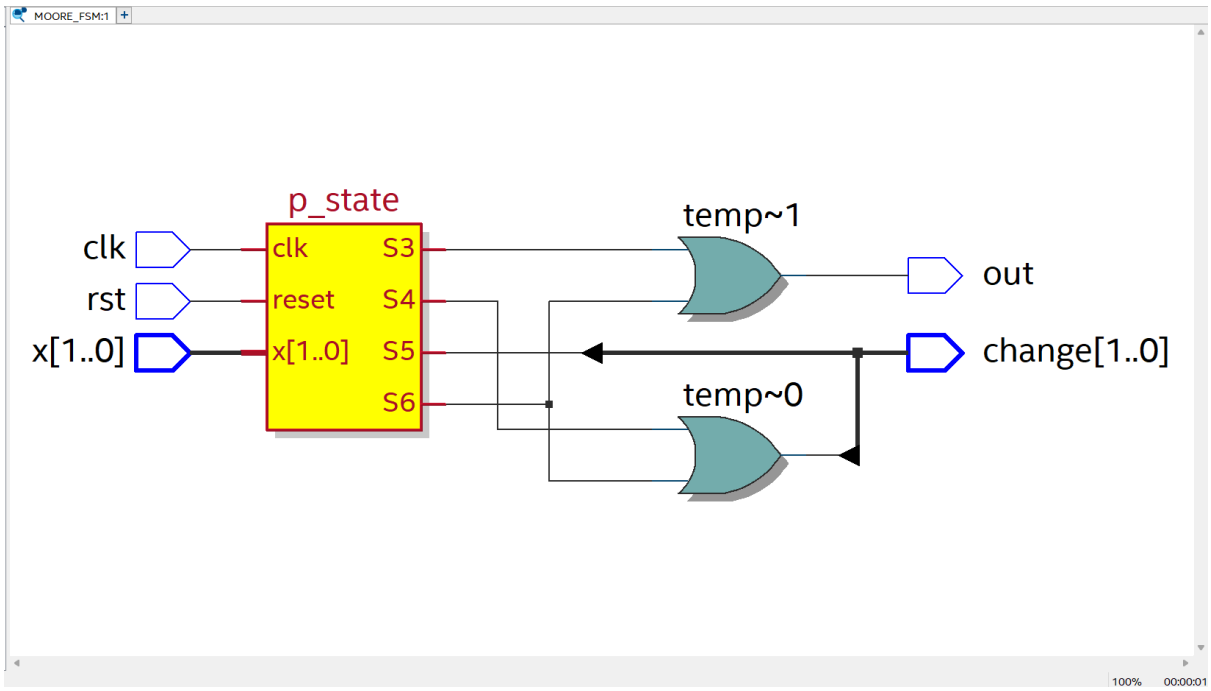

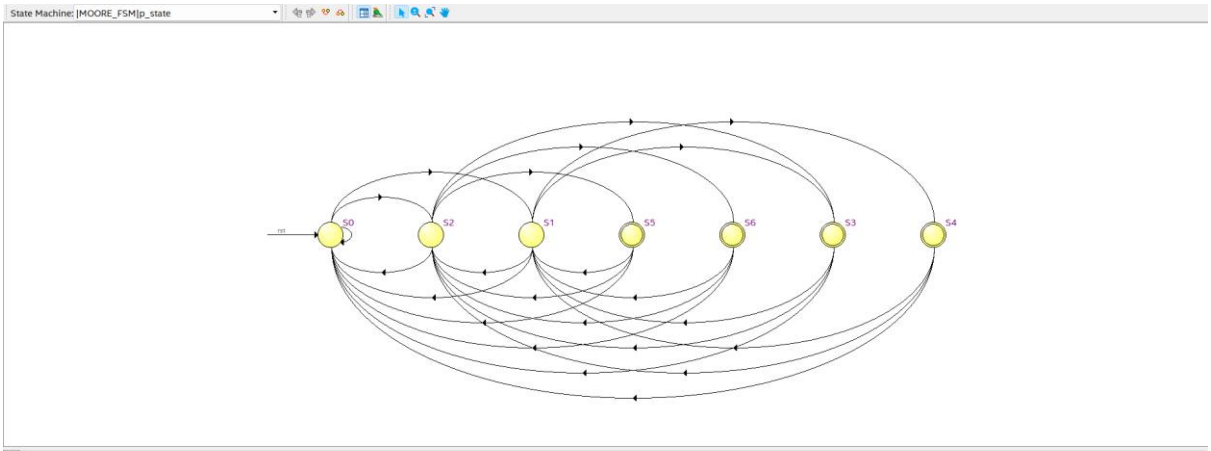

	Source State	Destination State	Condition
1	S0	S2	(!x[0]).(x[1])
2	S0	S1	(x[0]).(!x[1])
3	S0	S0	(!x[0]).(!x[1]) + (x[0]).(x[1])
4	S1	S4	(!x[0]).(!x[1])
5	S1	S3	(!x[0]).(x[1])

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SCREEN SHOTS OF CODE:

MEALY:

Text Editor - C:/Users/SANJAY ELAVARASAN/Desktop/FPGA/FSM_MEALY/MEALY_FSM - MEALY_FSM - [MEALY_FSM.v]

File Edit View Project Processing Tools Window Help

```
module MEALY_FSM(clk, rst, x, out, change);
  input clk, rst;
  input [1:0] x;
  output reg out;
  output reg [1:0] change;

  reg [1:0] p_state, n_state;

  parameter S0 = 2'b00;
  parameter S1 = 2'b01;
  parameter S2 = 2'b10;

  always @(p_state or x)
  begin
            n_state = S0;
          change= 2'b00;
case(p_state)
S0: if(x == 2'b01) begin //Received 5 Rs
  n_state = S1;
  change = 2'b00;
  out = 1'b0;
end
else if(x == 2'b10) begin   //Received 10 Rs
  n_state = S2;
  change = 2'b00;
  out = 1'b0;
end
else begin
  n_state = S0;
  change = 2'b00;
  out = 1'b0;
end
S1: if(x == 2'b01) begin //Received 10(5+ 5) Rs
  n_state = S2;
  change = 2'b00;
  out = 1'b0;
end
else if(x == 2'b10) begin             //Received 15(5+10) Rs
  n_state = S0;
  out = 1'b1;
  change = 2'b00;
end
else begin
  n_state = S0;
  change = 2'b01;
  out = 1'b0;
end
S2: if(x == 2'b01) begin //Received 15(5+ 5 + 5) Rs
  n_state = S0;
  out = 1'b1;
end
else if(x == 2'b10) begin //Received 20(10+ 10) Rs
  n_state = S0;
  out = 1'b1;
  change = 2'b01;
end
else begin
  n_state = S0;
  change = 2'b10;
  out = 1'b0;
end
endcase
end
 always @(posedge clk or posedge rst)
  if(rst)
    p_state <= S0;
        else
    p_state <= n_state;

 endmodule
```

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MOORE :

Text Editor - C:/Users/SANJAY ELAVARASAN/Desktop/FPGA/FSM_MEALY/MEALY_FSM - MEALY_FSM - [MOORE_FSM.v]

File Edit View Project Processing Tools Window Help

```
module MOORE_FSM(x,clk,rst,out,change);

input [1:0]x;
input rst;
input clk;
output out;
output [1:0]change;

reg [2:0] temp;

parameter S0 = 3'b000,
S1=3'b001,
S2=3'b010,
S3=3'b011,
S4=3'b100,
S5=3'b101,
S6=3'b110;

reg [2:0]p_state,n_state;

always @(posedge clk or posedge rst) begin
    if(rst)
       p_state <= S0;
    else
       p_state <= n_state;
end

always @(*) begin
    n_state = p_state;
    case(p_state)

    S0,S3,S4,S5,S6: begin
        if(x==2'b00) begin
            n_state = S0;
        end
        else if(x==2'b01) begin
            n_state = S1;
        end
        else if(x==2'b10) begin
            n_state = S2;
        end
        else begin
            n_state = S0;
        end
    end

    S1: begin
        if(x==2'b00) begin
            n_state = S4;

        end
        else if(x==2'b01) begin
```

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```
            n_state = S2;

        end
        else if(x==2'b10) begin
            n_state = S3;

        end
        else begin
            n_state = S0;

        end
    end

    S2: begin
        if(x==2'b00) begin
            n_state = S5;

        end
        else if(x==2'b01) begin
            n_state = S3;

        end
        else if(x==2'b10) begin
            n_state = S6;

        end
        else begin
            n_state = S0;
        end

    end

    default n_state = S0;


    endcase


end
```

```
//OUTPUT LOGIC
always @(*) begin
    case(p_state)
        S0: temp = 3'b000;
        S1: temp = 3'b000;
        S2: temp = 3'b000;
        S3: temp = 3'b001;
        S4: temp = 3'b010;
        S5: temp = 3'b100;
        S6: temp = 3'b011;
        default: temp = 3'b000;
    endcase
end

assign out = temp[0];
assign change = temp[2:1];


endmodule
```

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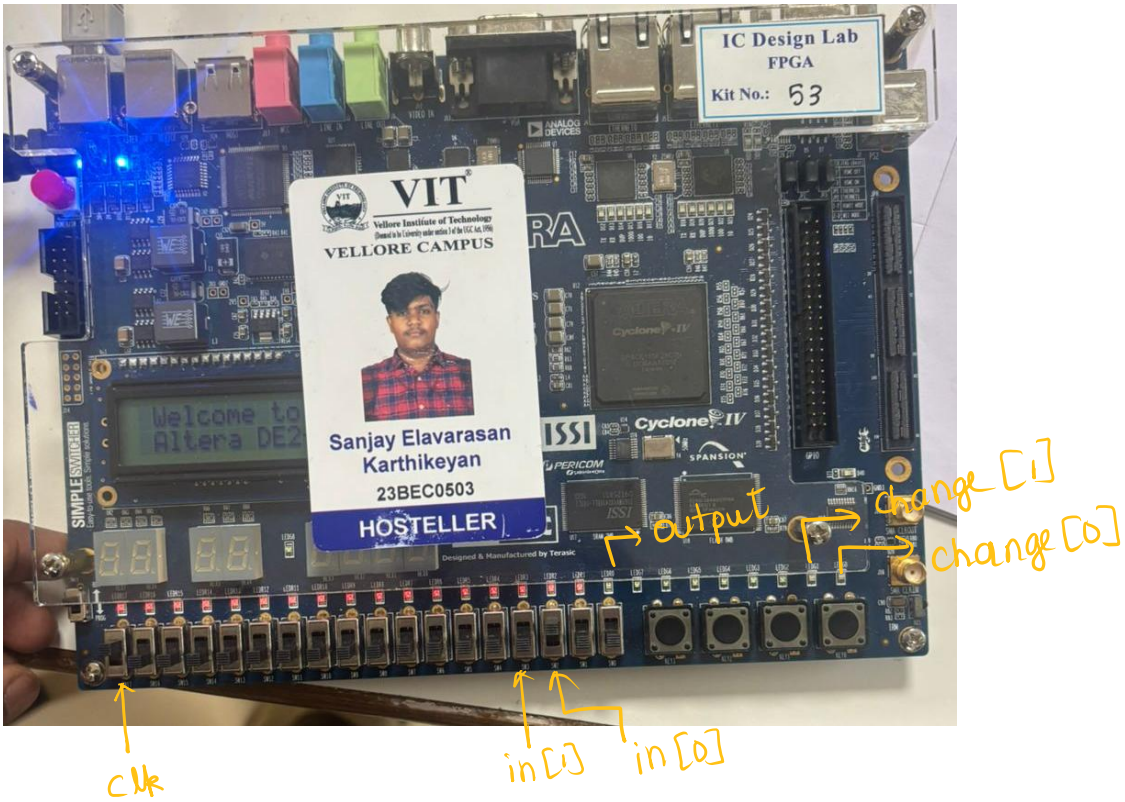
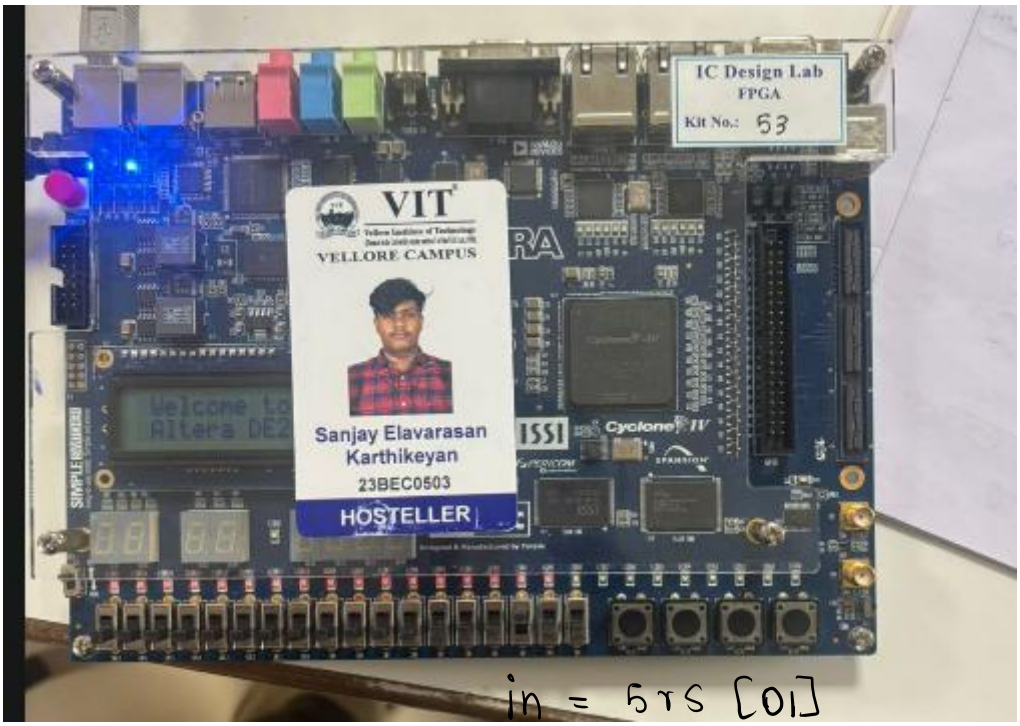
OUTPUT [FPGA]:

MEALY:

in = 5 rs [01]





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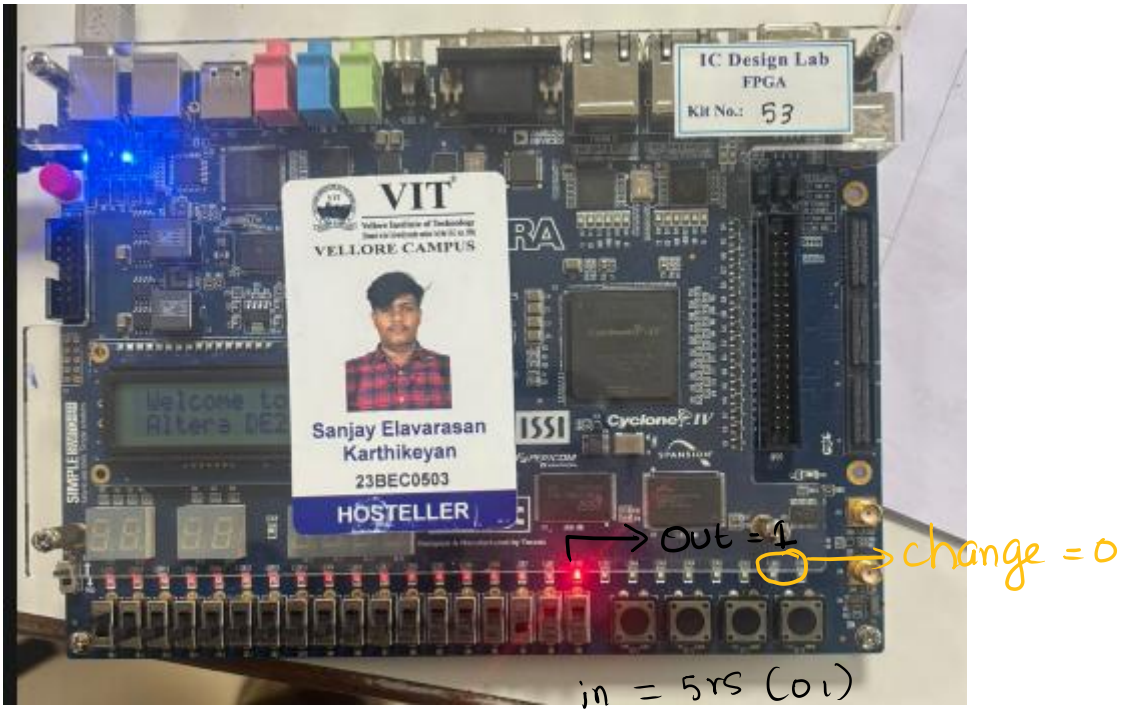
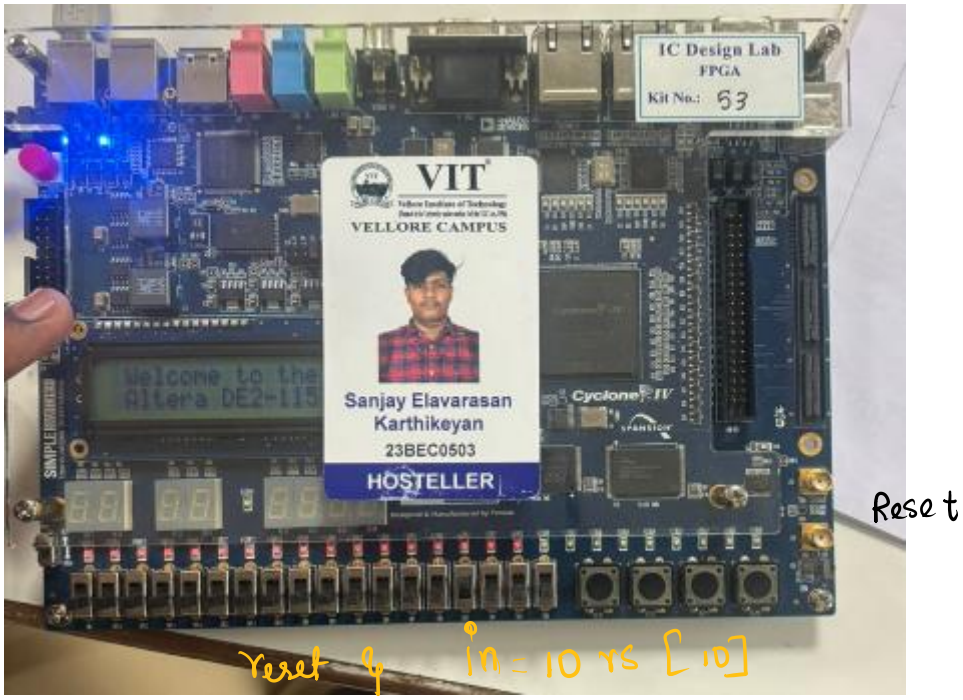

IC Design Lab
FPGA
Kit No.: 53
VIT
VELLORE CAMPUS
Sanjay Elavarasan
Karthikeyan
23BEC0503
HOSTELLER
out = 1
change = 0
in = 5rs (01)


IC Design Lab
FPGA
Kit No.: 53
VIT
VELLORE CAMPUS
Sanjay Elavarasan
Karthikeyan
23BEC0503
HOSTELLER
Reset
reset
in = 10 rs [10]

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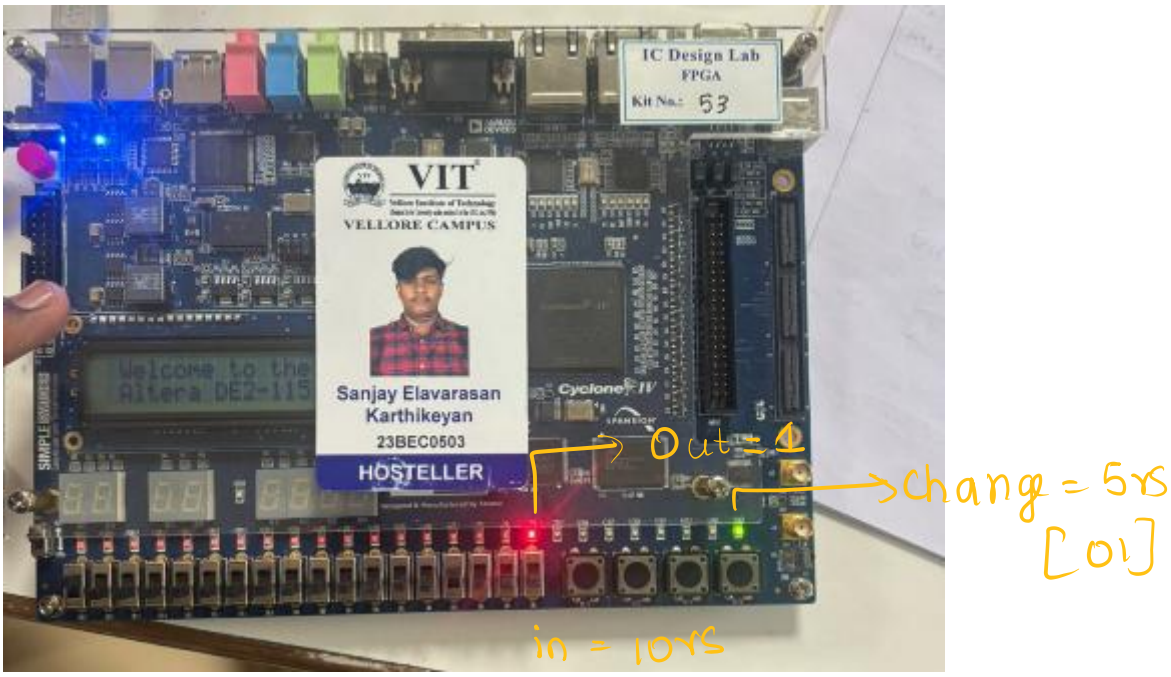
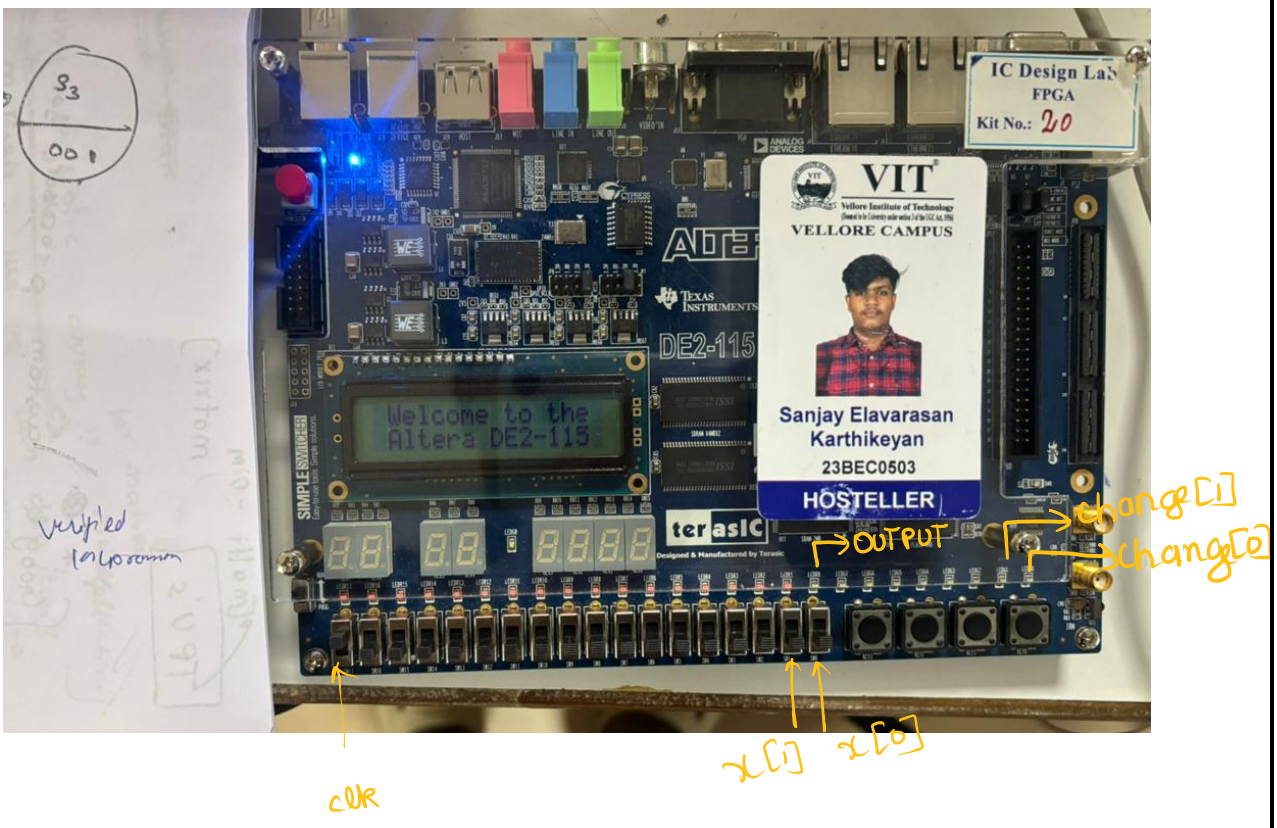


MOORE:



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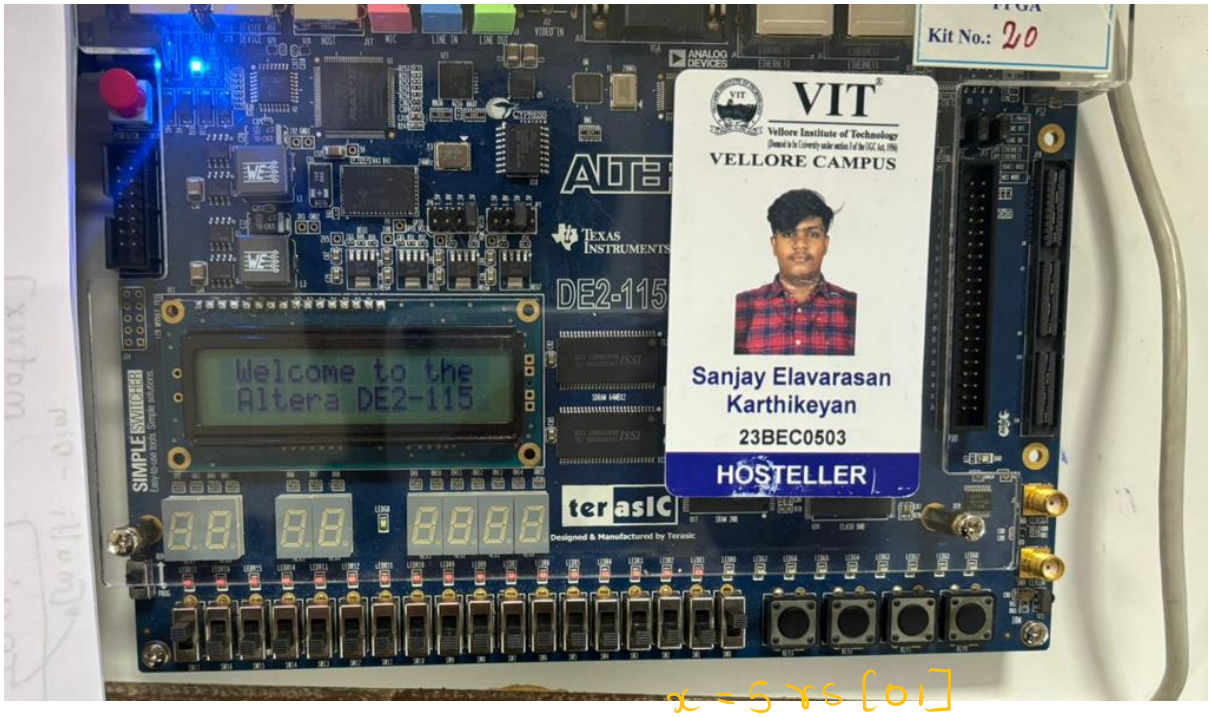
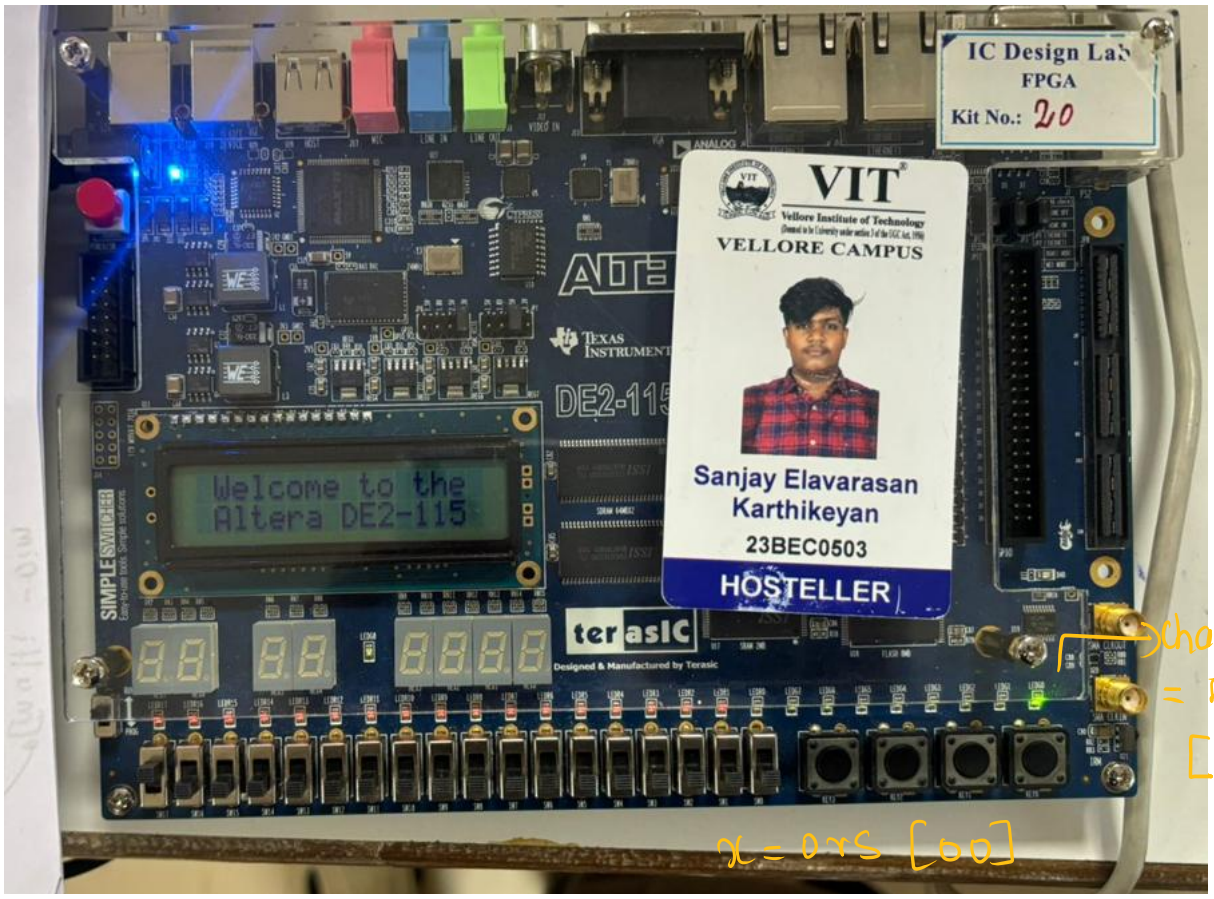

Kit No.: 20
VIT
Vellore Institute of Technology
VELLORE CAMPUS
Sanjay Elavarasan
Karthikeyan
23BEC0503
HOSTELLER
Welcome to the
Altera DE2-115
DE2-115
x = 5 ns [01]


IC Design Lab
FPGA
Kit No.: 20
VIT
Vellore Institute of Technology
VELLORE CAMPUS
Sanjay Elavarasan
Karthikeyan
23BEC0503
HOSTELLER
Welcome to the
Altera DE2-115
change
= 5ns
[01]
x = 0 ns [00]

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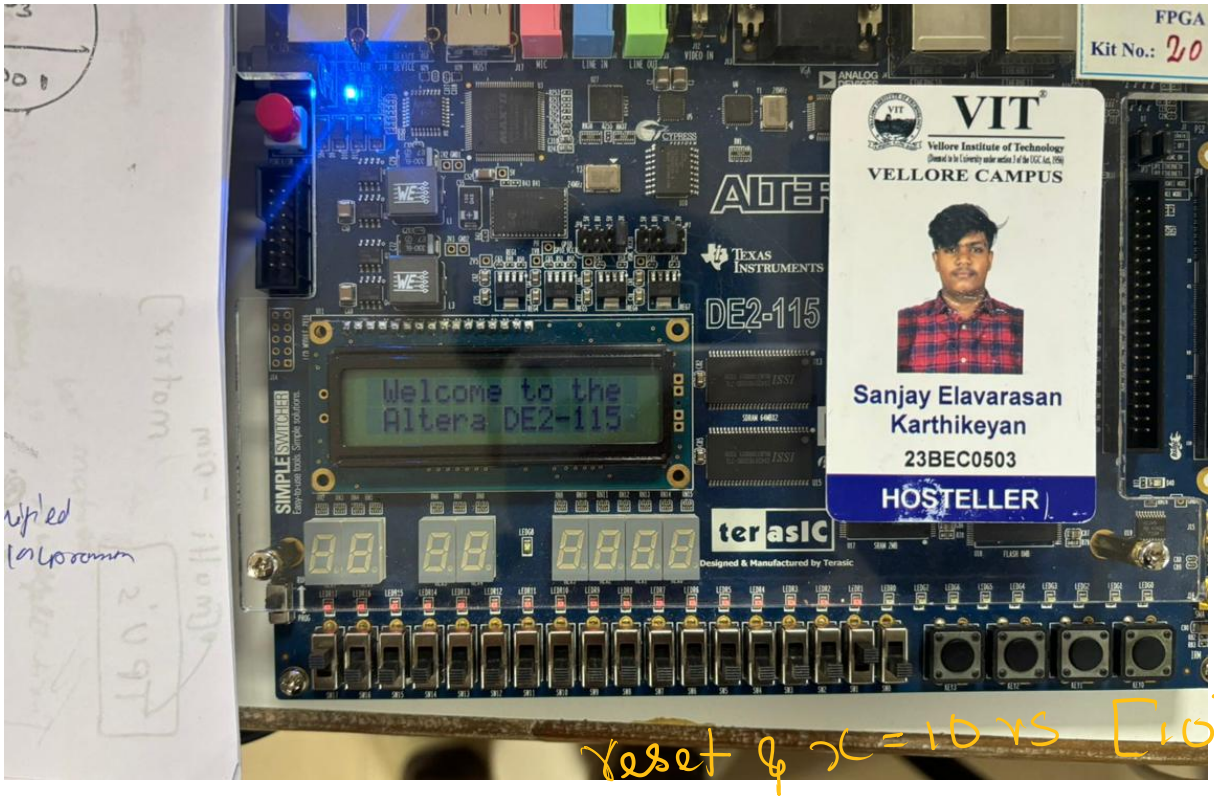
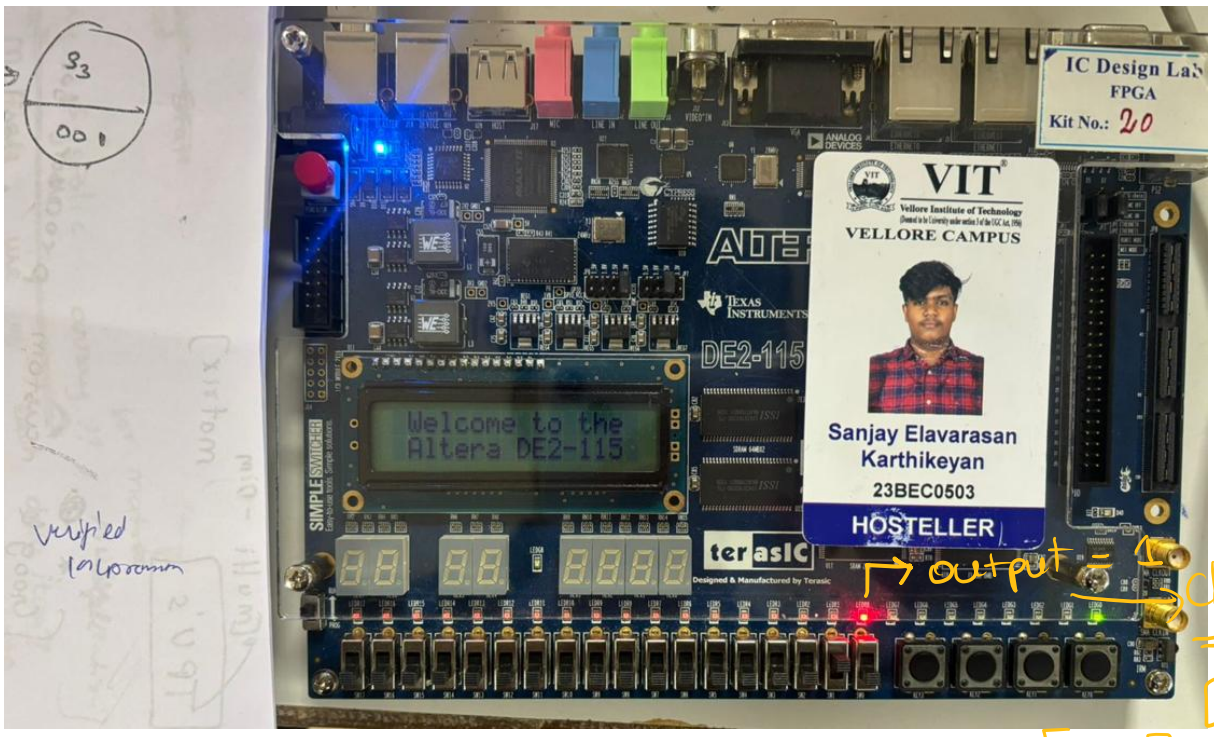


reset & x = 10 ns [10]



output = 1

change = 5 ns [01]

x = 10 ns [10]

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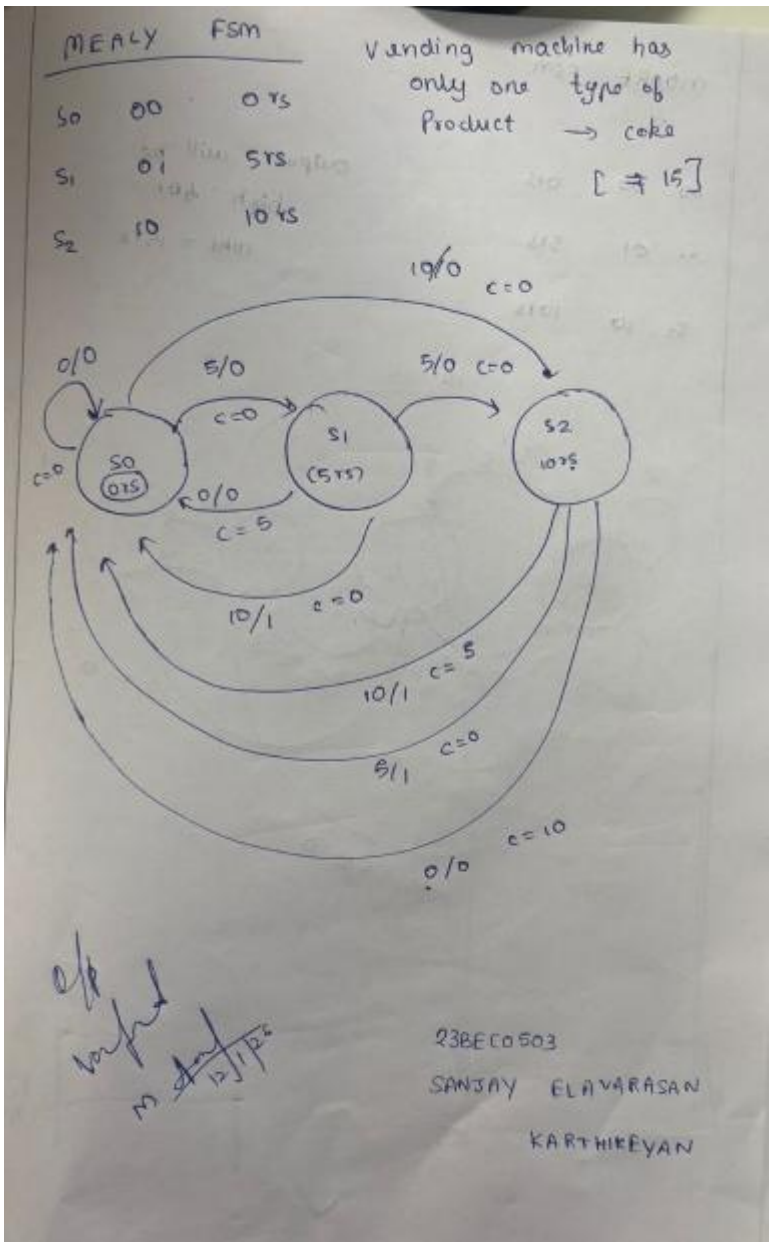
Attestation copy:

MEALY:



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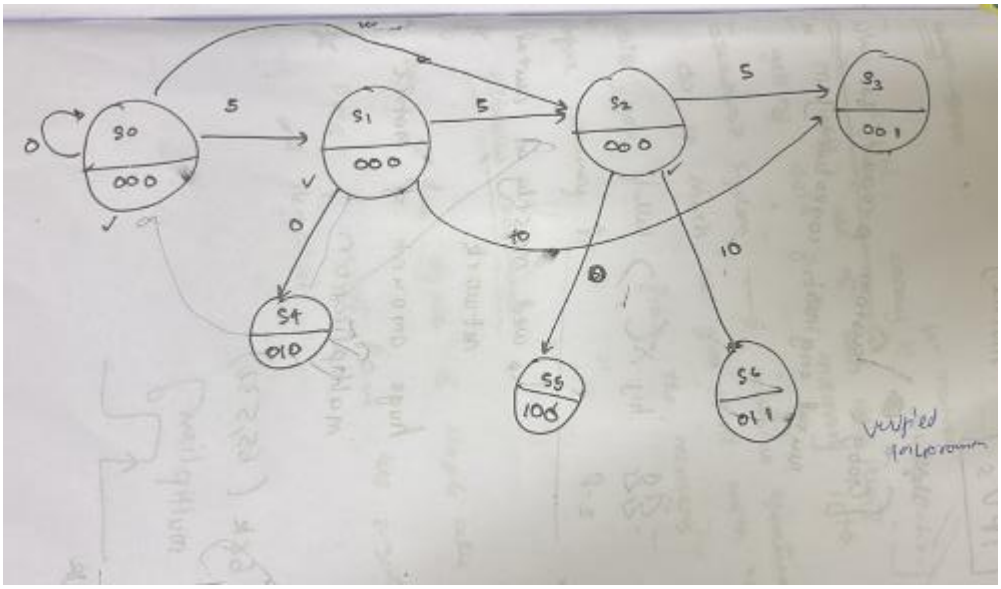
MOORE:



INFERENCE:

The Coke vending machine was successfully designed using Mealy and Moore FSM models. The system correctly accepts Rs.5 and Rs.10, dispenses Coke at Rs.15, and returns change when required. Simulation in ModelSim and implementation on the DE2-115 FPGA verified correct operation. This confirms that FSM-based design is effective for vending machine applications.